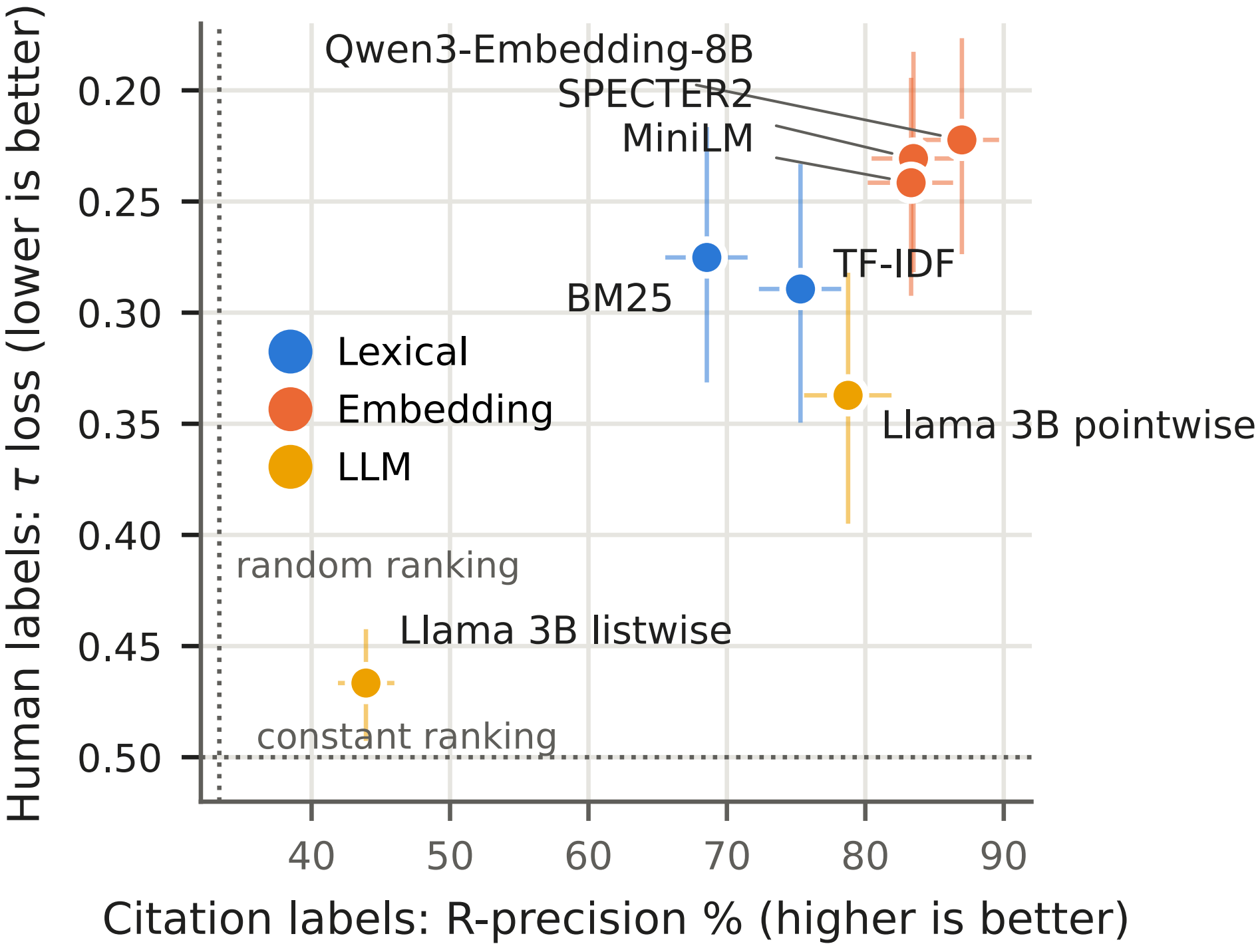
Qwen3-Embedding-8B
SPECTER2
MiniLM
BM25
TF-IDF
Llama 3B pointwise
Llama 3B listwise
Lexical
Embedding
LLM
random ranking
constant ranking
0.20
0.25
0.30
0.35
0.40
0.45
0.50
40
50
60
70
80
90
Human labels: τ loss (lower is better)
Citation labels: R-precision % (higher is better)